

Sistem Informasi Geografis



Dr. Sirojul Munir S.Si., M.Kom

STT Terpadu Nurul Fikri





Komponen Data SIG

Dr. Sirojul Munir S.Si.,M.Kom | rojulman@nurulfikri.ac.id



   @rojulman

Geographical Information System

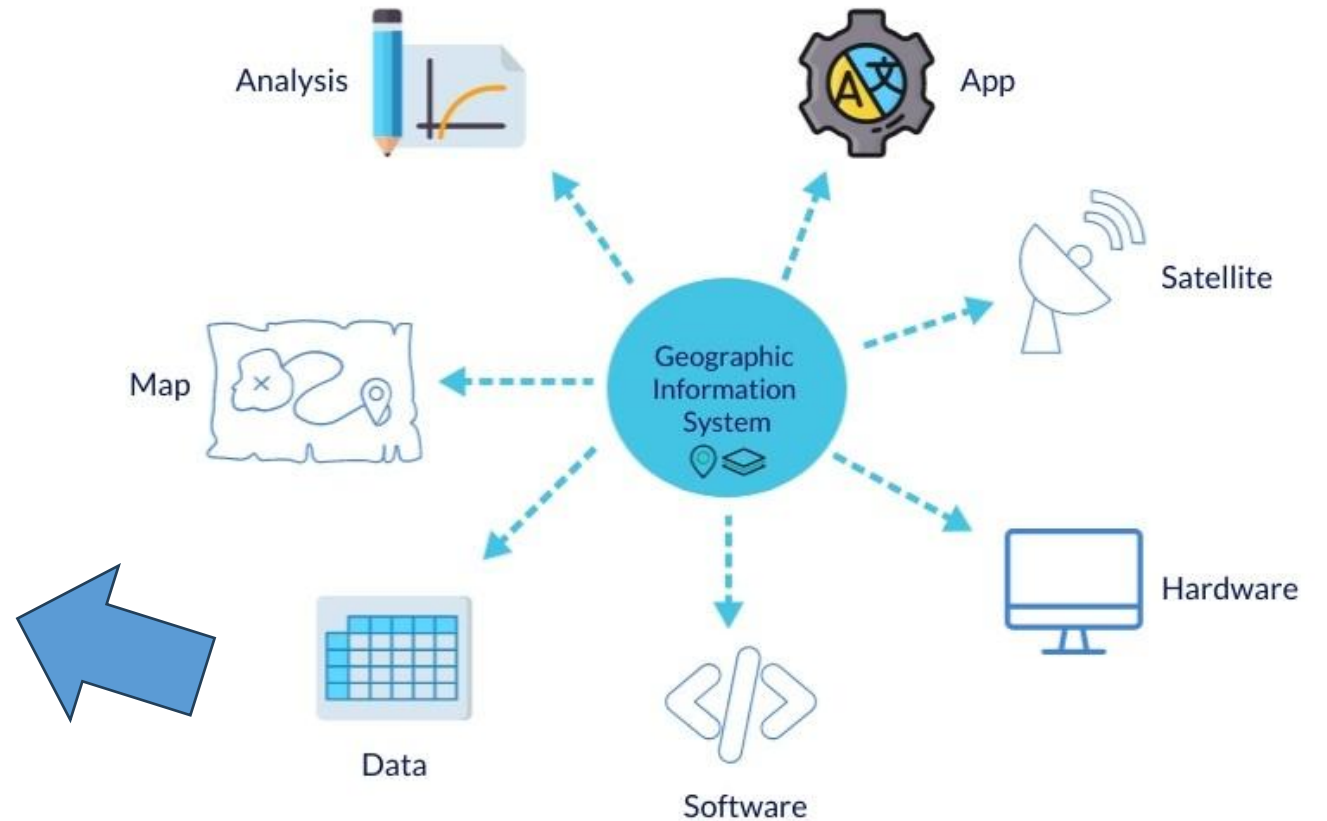
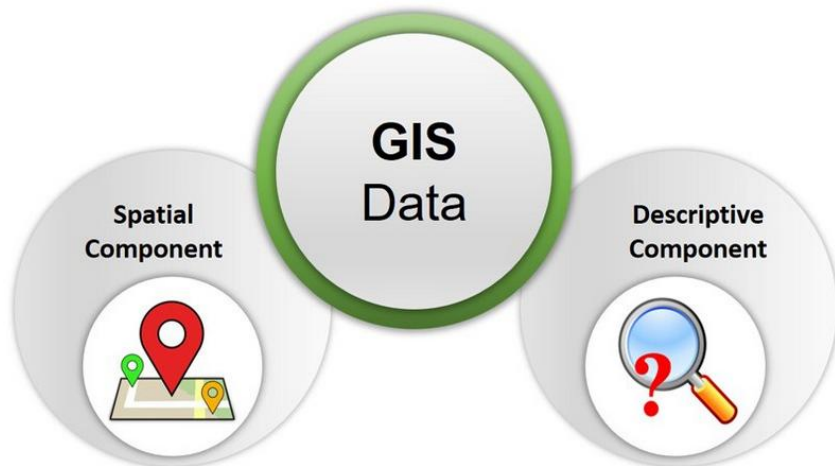


1. Hardware
2. Software
3. People
4. Data
5. Methods

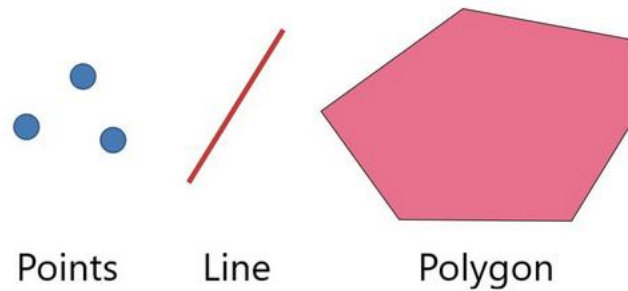


GIS Data Component

1. Data Spatial
2. Data Non Spatial
(*Descriptive*)



- Vector Data



- Raster Data

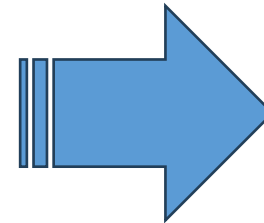
		Columns									
		1	2	3	4	5	6	7	8		
Rows	1									Rows	1
	2										2
	3						6, 3				3
	4			3, 4							4
	5										5
	6										6
	7										7
	8										8

Data Vector dan Raster dapat tersimpan dalam format SHAPEFILE bentuk ZIP berisi:

- Informasi fitur geometry (shp)
- Dataset semua fitur atribut (.dbf)
- Index spatial (.shx)
- Informasi project dan sistem koordinat (.prj)
- Diskripsi encoding shapefile (.cpg)
- Optional file spatial index lainnya (.sbn, .sbx)

Vector Data :: GeoJSON data

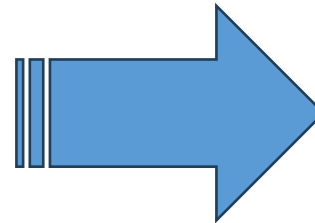
- GeoJSON: Format data berbasis JSON (JavaScript Object Notation) yang didesain untuk mewakili fitur geografis seperti luasan spatial dan propertinya serta atribut tambahan data non-spatial
- Object GeoJSON dapat menunjukan geometri (Point, LineString, Polygon)
- GeoJSON dapat sebagai inputan data pada aplikasi Qgis/Arcgis
- GeoJSON dapat diimplementasi untuk peta pada aplikasi berbasis web



	</>JSON	Table
1	{	
2	"type": "FeatureCollection",	
3	"features": [	
4	{	
5	"type": "Feature",	
6	"properties": {	
7	"lota": "Jakarta",	
8	"luas": 661.5	
9	},	
10	"geometry": {	
11	"coordinates": [	
12	106.82668996573705,	
13	-6.175851454714007	
14	],	
15	"type": "Point"	
16	}	
17	}	
18	]	
19	}	

Vector Data :: GeoJSON data

- Feature Collection:
daftar list fitur dari object
- Geometry Object Type:
point, line, polygon dengan gps koordinatnya
- Features Object:
data yang berasosiasi dengan objek geometry



</>JSON		Table
1	{	
2	"type": "FeatureCollection",	
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4	{	
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12	106.82668996573705,	
13	-6.175851454714007	
14	],	
15	"type": "Point"	
16	}	
17	}	
18	]	
19	}	

geojson.io powered by mapbox [Sign up for Mapbox](#)

Open Save New Meta

Search

1 {
2 "type": "FeatureCollection",
3 "features": []
4 }

Globe Mercator

Standard Satellite Streets Outdoors Light Dark OSM

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[geojson.io/#map=14.15/-6.17591/106.82619](#)

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Standard

Satellite

Streets

Outdoors

Light

Dark

OSM

Tanah Abang

mapbox

1

{

2

"type": "FeatureCollection",

3

"features": [

4

{

5

"type": "Feature",

6

"properties": {},

7

"geometry": {

8

"coordinates": [

9

[

10

[

11

106.82333137024733,

12

-6.171490562842521

13

],

14

[

15

106.82308890853875,

16

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17

],

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[

19

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-6.179862942028194

21

],

22

[

23

106.82338343193544,

24

-6.180350961604887

25

],

26

],

27

}

28

}

29

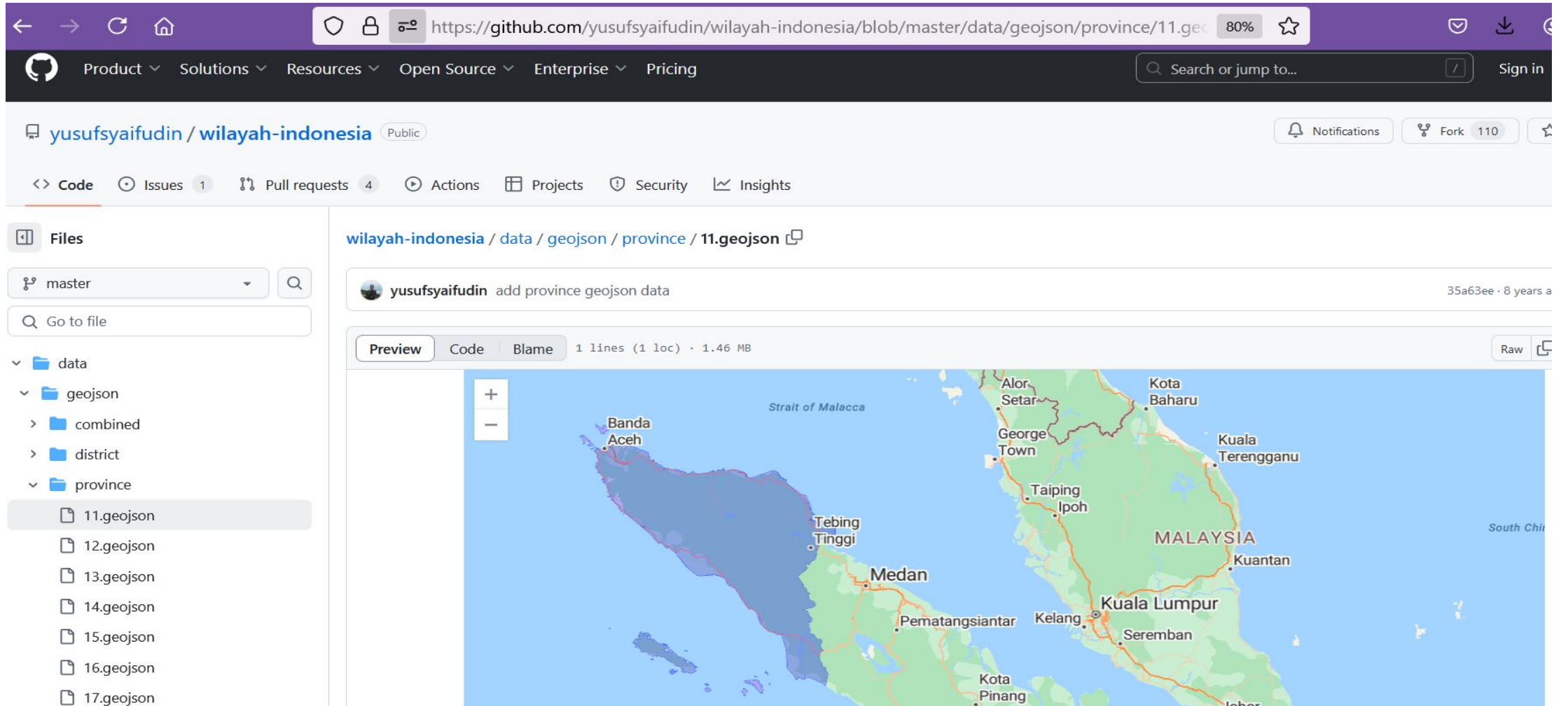
}

30

}

© Mapbox © OpenStreetMap Improve this map

GeoJSON data spatial :: polygon area



The screenshot displays a GitHub repository page for 'yusufsyaifudin / wilayah-indonesia'. The browser address bar shows the URL: `https://github.com/yusufsyaifudin/wilayah-indonesia/blob/master/data/geojson/province/11.geojson`. The repository page includes navigation tabs for Code, Issues (1), Pull requests (4), Actions, Projects, Security, and Insights. The left sidebar shows the file structure: `data` (expanded) contains `geojson` (expanded), which contains `combined`, `district`, and `province` (expanded). Under `province`, files `11.geojson` through `17.geojson` are listed. The main content area shows the file `11.geojson` with a preview of a map of Indonesia. A blue polygon is overlaid on the map, highlighting a specific region in the western part of the country. The map includes labels for various cities and regions, such as Banda Aceh, Medan, Kuala Lumpur, and Kota Baharu. The map also shows the Strait of Malacca and the South China Sea.

kodewilayah.id

Referensi lengkap nama dan kode wilayah se-Indonesia

Cari

Telusuri berdasarkan provinsi

11 Aceh

12 Sumatera Utara

13 Sumatera Barat

16 Sumatera Selatan

14 Riau

21 Kepulauan Riau

15 Jambi

17 Bengkulu

18 Lampung

19 Bangka Belitung

31 DKI Jakarta

36 Banten

32 Jawa Barat

33 Jawa Tengah

34 DI Yogyakarta

35 Jawa Timur

51 Bali

61 Kalimantan Barat

62 Kalimantan Tengah

63 Kalimantan Selatan

64 Kalimantan Timur

52 Nusa Tenggara Barat

53 Nusa Tenggara Timur

71 Sulawesi Utara

72 Sulawesi Tengah

73 Sulawesi Selatan

74 Sulawesi Tenggara

75 Gorontalo

76 Sulawesi Barat

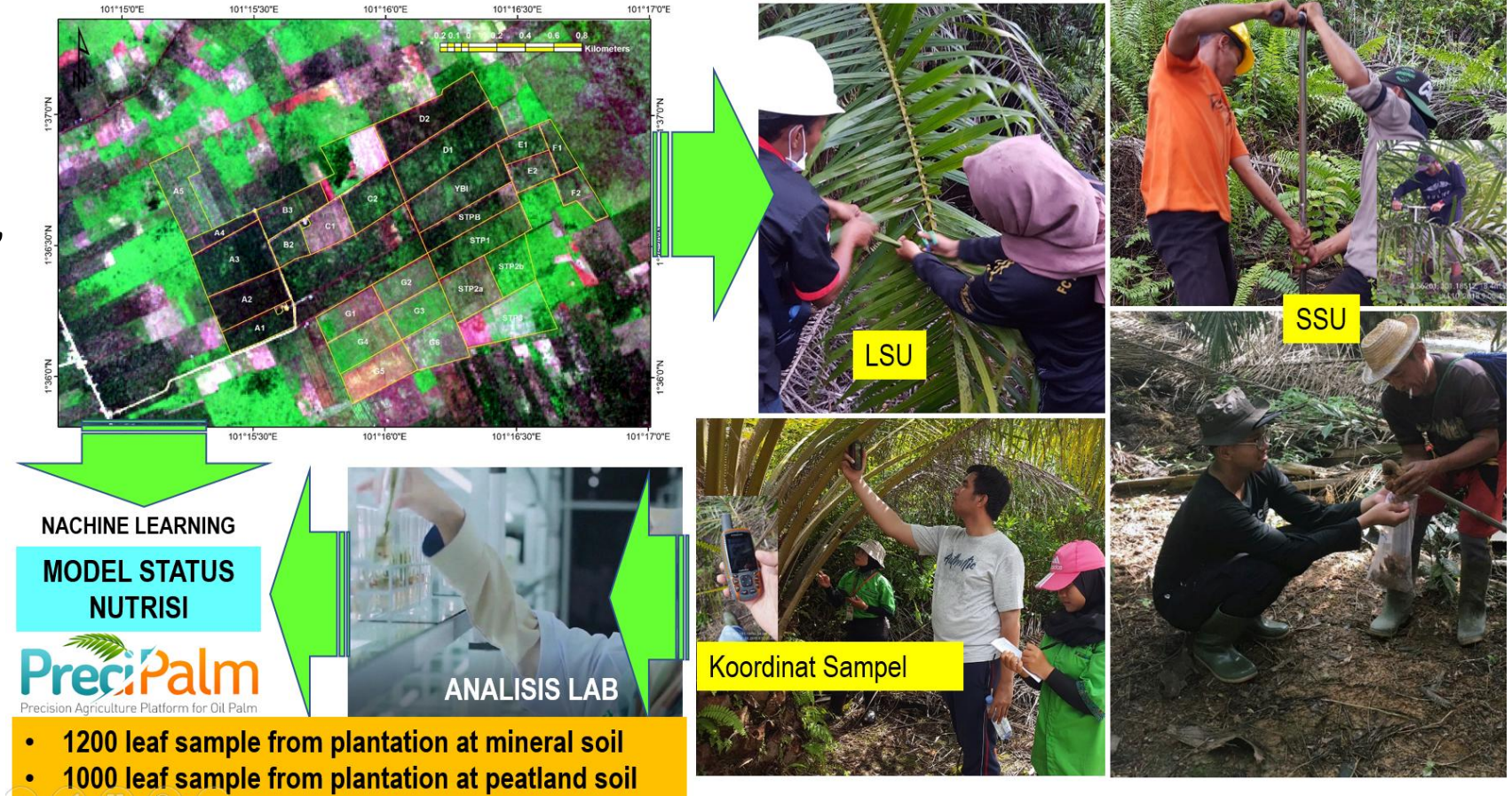
81 Maluku

82 Maluku Utara

91 Papua

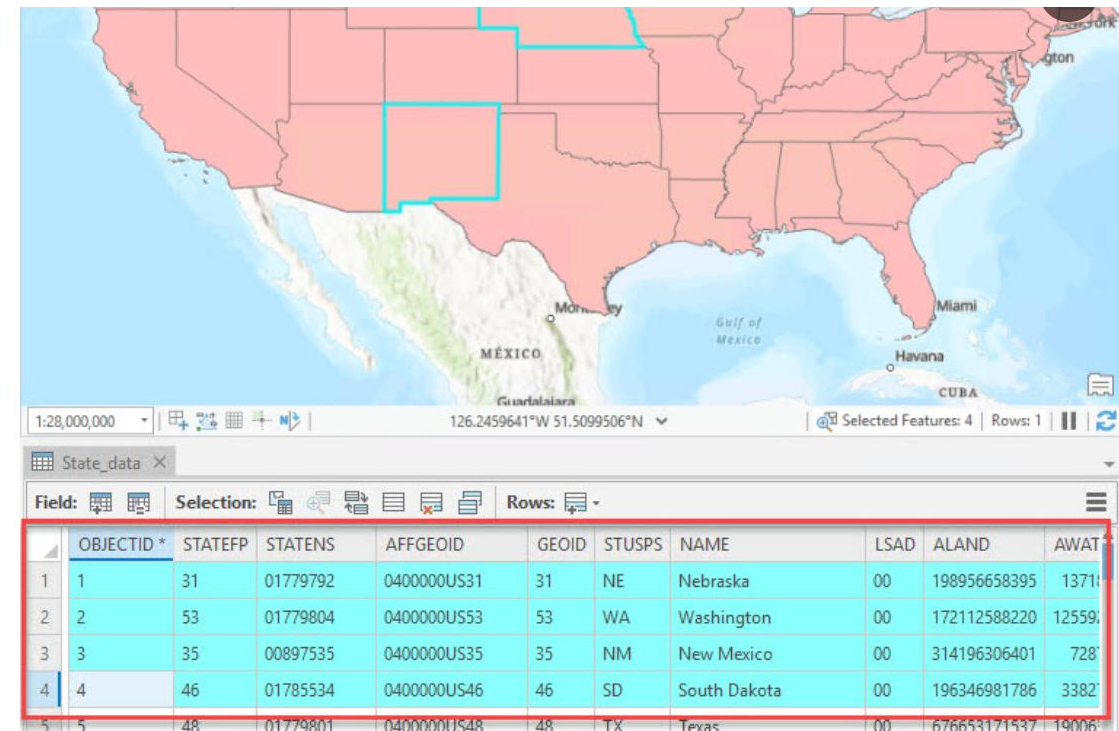
1

Data hasil pengukuran lapangan: data batas administrasi, batas kepemilikan lahan, batas hak pengusahaan hutan, hasil analisis lab nutrisi sampel tanah (SSU) dan daun tanaman (LSU)



2 Data disajikan dalam bentuk atribut referensi berbentuk tabel (data non grafis)

No.	Kecamatan	Jumlah Penduduk	LUas Wilayah	Kepadatan



Sumber data GIS :: data non spatial

2 Data disajikan dalam bentuk atribut referensi berbentuk tabel (data non grafis)

Car ID	Departure			Arrival		
	Date-Time	Latitude	Longitude	Date-Time	Latitude	Longitude
...	...	...	...	...	...	...
...	...	...	...	...	...	...
...	...	...	...	...	...	...



Sumber data GIS :: data non spatial

<https://www.bps.go.id/id/query-builder>

2



The screenshot shows the BPS (Badan Pusat Statistik) website interface. The breadcrumb trail is: Beranda > Produk - Tabel Statistik > Kependudukan dan Migrasi > Jumlah Penduduk Pertengahan Tahun. The left sidebar lists various statistical categories, with 'Kependudukan dan Migrasi' selected. The main content area displays the title 'Jumlah Penduduk Pertengahan Tahun (Ribu Jiwa), 2022-2024' and the update date 'Terakhir Diperbarui : 28 Juni 2024'. Below this are buttons for 'Kembali', 'Unduh', '</> JSON', and 'Bagikan'. There are also buttons for different time periods: '2022-2024' (selected), '2019-2021', '2016-2018', and '1960-2015'. A 'Freeze judul kolom' toggle is present. At the bottom, a table shows the population data for Indonesia for the years 2022, 2023, and 2024.

Indonesia	Jumlah Penduduk Pertengahan Tahun (Ribu Jiwa)		
	2022	2023	2024
Indonesia	275.773,8	278.696,2	281.603,8

<https://www.bps.go.id/id/statistics-table/2/MTk3NSMy/jumlah-penduduk-pertengahan-tahun--ribu-jiwa-.html>

- ✓ Menggunakan aplikasi Quantum Gis dengan dataset bersumber dari data spatial GeoJSON
- ✓ Menambahkan data non spatial pada dataset GeoJSON